Engine Model(s): PT6A

73-10-01

OIL-TO-FUEL HEATER - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00, STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, 0.025 inch diameter, and will not be called out in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC01-001	Fuel, Engine
PWC03-001	Oil, Engine Lubricating
PWC05-061	Cloth, Abrasive Coated
PWC11-021	DELETED
PWC11-021C	Compound, Cold Carbon Removing
PWC11-027	Solvent, Petroleum
PWC11-031	Cleaner, Engine

3. Special Tools

Not Applicable

4. Fixtures, Equipment and Supplier Tools

Not Applicable

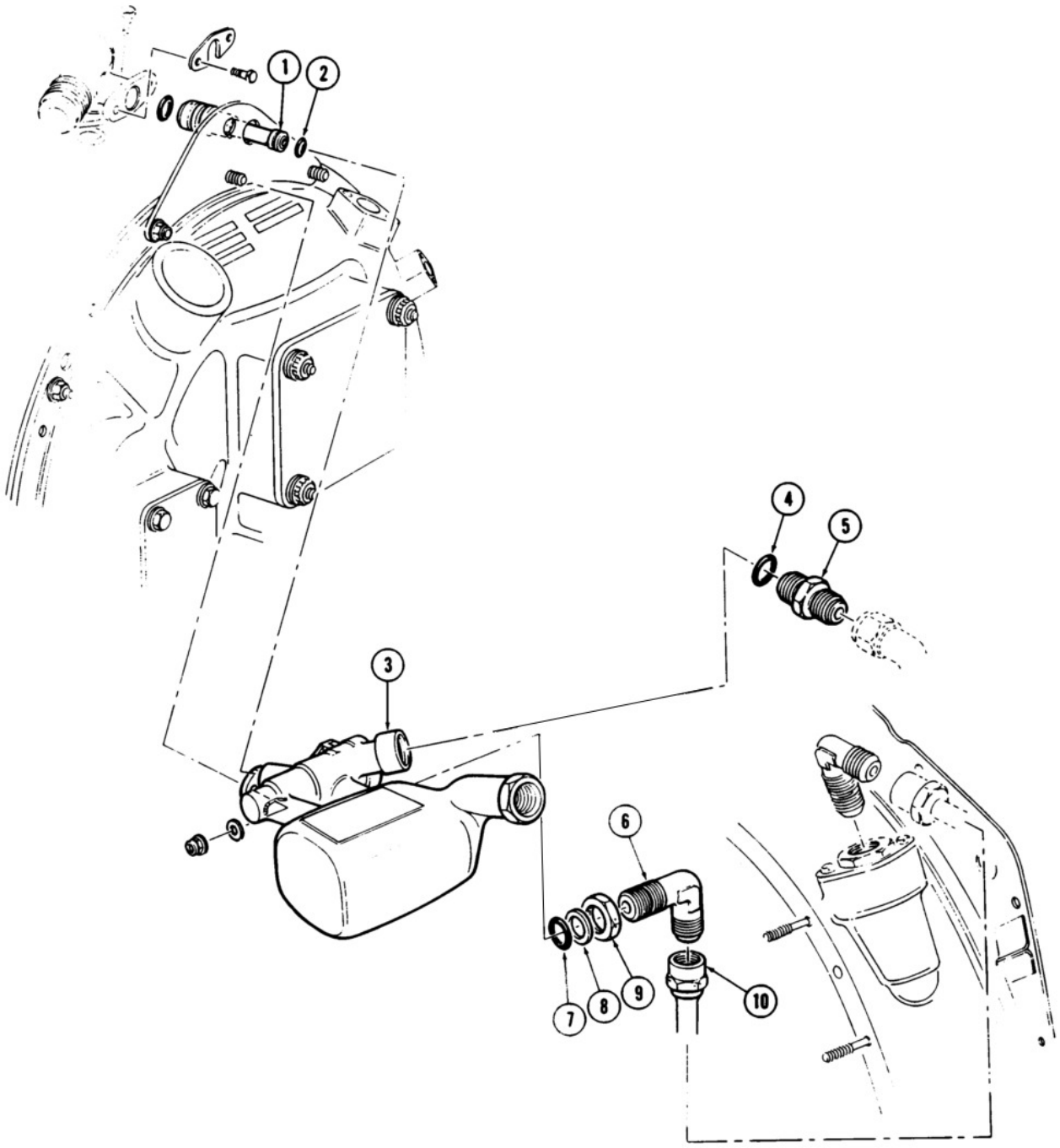
5. Removal/Installation

CAUTION: OBSERVE ALL FIRE AND SAFETY PRECAUTIONS.

A. Removal of Oil-to-Fuel Heater (Ref. Fig. 201)

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Figure 201 Removal/Installation of Oil-to-Fuel Heater



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1. Oil Return Check Valve
2. Preformed Packing
3. Oil-to-fuel Heater
4. Preformed Packing
5. Straight Nipple
6. Elbow
7. Preformed Packing
8. Back-up Ring
9. Locknut
10. Oil Inlet Tube

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(1) Drain oil tank (Ref. 72-00-00, SERVICING).

(2) Provide suitable drip pans for residual oil and fuel spillage.

(3) Disconnect the following lines from oil-to-fuel heater (3):

(a) Fuel inlet line (Ref.to airframe manufacturer's manual).

(b) Fuel outlet flexible hose from straight nipple (5).

(c) Oil inlet tube (10) from elbow (6).

(d) Cap all fuel lines.

(4) Remove oil tube (10) by disconnecting from check valve elbow on air inlet case.

(5) Remove two self-locking nuts and washers, and withdraw oil-to-fuel heater (3) from studs at flange G and from check valve (1).

(6) Remove preformed packing (2) from check valve (1). Suitably cap open port of valve.

(7) If oil-to-fuel heater is to be replaced, remove parts for reassembly on replacement unit:

(a) Airframe-supplied fitting at fuel inlet port.

(b) Elbow (6) and locknut (9). Discard preformed packing (7) and back-up ring (8).

(c) Straight nipple (5). Discard preformed packing (4).

B. Installation of Oil-to-Fuel Heater (Ref. Fig. 201)

(1) If a replacement oil-to-fuel heater (3) is being installed, depreserve prior to installation:

(a) Remove shipping plugs from oil ports and flush with engine oil (PWC03-001). Drain off surplus oil and reinstall shipping plugs until unit is required for installation.

(b) Remove shipping plugs from fuel ports and flush with engine fuel (PWC01-001). Drain off surplus fluid and reinstall shipping plugs until unit is required for installation.

(2) If a new oil-to-fuel heater is to be installed, assemble parts removed at disassembly:

(a) Depreserve unit as detailed in Subparagraph B.(1) .

(b) Install airframe-supplied fitting in fuel inlet port. Tighten and torque to airframe manufacturer's recommendation

(c) Install straight nipple (5) with preformed packing (4). Tighten and torque to 200 to 225 lb.in.

(d) Install elbow (6) with locknut (9), back-up ring (8) and preformed packing (7) (Ref. 70-00-00, REMOVAL/INSTALLATION). Do not final torque locknut (9) until angle of elbow has been established. (Ref. Subpara. (5) , following.)

(3) Install preformed packing (2) on oil return check valve (1).

(4) Locate oil-to-fuel heater (3) over studs at flange G and mate with oil return check valve (1). Secure with two washers and self-locking nuts. Tighten nuts and torque to 32 to 36 lb.in.

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(5) Connect oil inlet tube (10) to check valve elbow on inlet case and elbow (6) on heater unit:

(a) Final torque locknut (9) on elbow (6) to 70 to 80 lb.in.

(b) Tighten tube coupling nut at elbow (6) and torque to 270 to 300 lb.in.

(c) Tighten tube coupling nut at check valve elbow and torque to 90 to 100 lb.in.

(d) Lockwire coupling nuts and locknut.

(6) Connect fuel hose to straight nipple (5). Tighten coupling nut, torque to 450 to 500 lb.in., and lockwire.

(7) Connect airframe fuel supply line to fitting at inlet port. Tighten coupling nut and torque to airframe manufacturer's recommendation, and lockwire.

(8) Replenish oil tank (Ref. 72-00-00, SERVICING).

6. Cleaning/Painting

A. Cleaning

(1) Soak the external oil tube in the cold carbon remover (PWC11-021C) for two hours at room temperature.

(2) Remove the tube from the solvent and pull a suitable sized swab (or lint-free cloth) through the tube.

CAUTION: MAKE SURE NO PART OF THE SWAB OR CLOTH REMAINS IN THE OIL TUBE; OTHERWISE, THE OIL FLOW CAN BE RESTRICTED.

(3) Rinse the tube exterior with petroleum solvent (PWC11-027) or (PWC11-031).

(4) Pressure flush the tube interior with petroleum solvent and dry the interior and exterior with compressed air.

(5) Clean all components of the oil return check valve assembly with petroleum solvent (PWC11-027) or (PWC11-031). Make sure the check valve, spring, the spring seat and interior of the check valve housing are completely clean.

7. Inspection/Check

A. Oil-to-fuel Heater

(1) Visually inspect the mounting lugs for cracks and other defects. Cracks are not permissible; minor defects may be repaired.

(2) Inspect the bores of the oil inlet and outlet ports and the fuel outlet port for damage on the sealing surfaces.

(3) Inspect the fuel inlet port for damage to the threads.

(4) Check for security of the identification plate.

(5) Oil inlet tube:

(a) Inspect the tube for damage.

(b) Cracks are not repairable.

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(c) A dented tube must be replaced if any of the following conditions apply:

1 Sharp cornered dent.

2 Length and depth of the dent exceeds 10 percent of the tube outside diameter.

3 More than one dent per 12-inch length.

4 If the dent is within one inch of the ferrule scarf-welds or bend.

(d) Scratches, Nicks or Chafing. Light scratching of no appreciable depth can be ignored. Tubes must be replaced if scratches, nicks or chafing cannot be blended out within an area of one-half square inch to a maximum depth of one quarter of the original tube wall thickness. Any number of such blended areas are permitted, provided they are separated by at least one-half inch of undamaged material.

(e) Pitting. Tubes must be replaced if pitting cannot be blended out within an area of one-half square inch to a maximum depth of one quarter of the original tube wall thickness. Any number of such blended areas are permitted, provided they are separated by at least one-half inch of undamaged material.

(f) Corrosion. Rust and stains are acceptable if they can be removed by light polishing with a crocus cloth; otherwise, Step (e) applies.

8. Approved Repairs

A. Oil-to-Fuel Heater

(1) Minor damage such as burrs, scores, scratches, nicks and other similar defects on the sealing surfaces of oil and fuel ports may be cleaned up by blending with a fine grade crocus cloth (PWC05-061), making sure that all high spots are removed.

(2) Blend out minor surface defects on the mounting lugs and other metal surfaces with a fine stone or swiss file.

(3) Clean up minor thread damage in the fuel and oil ports or chase the threads.

9. Adjustment/Test

A. Oil-To-Fuel Heater

(1) Start engine (Ref. 71-00-00, ADJUSTMENT/TEST, Engine Starting) and do checks associated with Installation/Replacement of Oil-to-fuel Heater (Ref. 71-00-00, ADJUSTMENT/TEST, Table 501).

NOTE: Run engine above 72 % (Ng) until oil temperature stabilizes

(2) Check the oil-to-fuel heater within 15 min. after engine shutdown for a high operating temperature. A heater fuel outlet or a fuel pump filter cover having a temperature above 140°F. (60°C.) after shutdown shows a heater that does not correctly operate. To measure the temperature of the oil-to-fuel heater, do one of the following procedures:

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WARNING: MAKE SURE PROTECTIVE CLOTHING IS PUT ON WHEN MANUALLY CHECKING THE TEMPERATURE OF THE HEATER FUEL OUTLET OR THE FUEL PUMP FILTER COVER. THE OIL-TO-FUEL HEATER CAN GET HOT. THIS WILL PREVENT POSSIBLE INJURY TO PERSONNEL.

(a) Carefully feel the heater fuel outlet or the fuel pump filter cover.

1 If the temperature of the heater fuel outlet or the fuel pump filter cover feels unusually hot, replace the oil-to-fuel heater.

2 If a high temperature is not felt, remove the protective clothing and carefully check the heater fuel outlet or the fuel pump filter cover again. If the oil-to-fuel heater is operating correctly, the heater fuel outlet and the fuel pump filter cover should be comfortable to touch.

CAUTION: A NEW TEMPERATURE RECORDER MUST BE USED FOR EACH INSPECTION.

(b) Put a temperature recorder on the heater fuel outlet or the fuel pump filter cover. If the temperature recorder shows a temperature above 140°F. (60°C.), replace the oil-to-fuel heater.

NOTE: A temp-plate temperature recorder (PWC05-329) or similar can be used.

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